

Devi Ahilya University
Institute of Engineering & Technology, Indore
Class Test: III

Sub Code: 3ETRG1/3EIRG1

Class: II Year

Duration: 70 Min.

Sub Name: Network Analysis

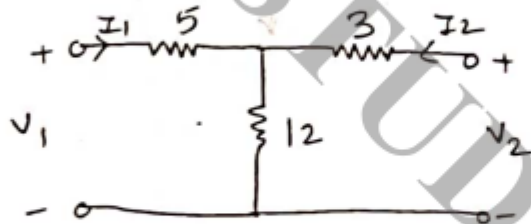
Branch: E & TC (A and B Section)/EI

Maximum Marks: 20

Note: All questions carry equal marks.

Q.1 Calculate the Z and Y parameters for the network given in fig. 1.

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Q.2 Solve the differential equation $\frac{d^2V}{dt^2} + 5\frac{dV}{dt} + 6V = e^{-2t} + 5e^{-3t}$ for $V(0+) = 1$ and $\frac{dV}{dt}(0+) = -1$

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Q.3 Check the function for positive realness $F(s) = \frac{s+4}{s^2+2s+5}$

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Q.4 Synthesis the network using Cauer I and II form if $Z(s) = \frac{s^4+7s^2+9}{s(s^2+4)}$

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